

Emma Andrews

✉ e.andrews@ufl.edu | 📧 emma-andrews | 🌐 emma-andrews- | L3Harris Endowment Ph.D. Fellow

Education

University of Florida

PH.D. IN COMPUTER SCIENCE

Gainesville, Florida

May 2023 - May 2027

University of Florida

M.S. IN COMPUTER SCIENCE, GPA: 3.88

Gainesville, Florida

Aug. 2021 - May 2023

University of Florida

B.S. IN COMPUTER SCIENCE WITH MATHEMATICS MINOR, GPA: 3.92

Gainesville, Florida

Aug. 2017 - May 2021

Skills

Programming Python, C/C++, Julia, Rust, Java, Verilog

Technologies Windows, Linux, Git, VxWorks

Professional Experience

University of Florida CISE Department

GRADUATE RESEARCH ASSISTANT, EMBEDDED SYSTEMS LAB

Gainesville, Florida

Aug. 2022 - Present

- Researching hardware-optimized tensor decompositions for machine learning and quantum computing applications
- Exploring large-scale classical simulation of quantum many-body systems using AI-generated tensor decompositions
- Designed a framework to choose optimal tensor decompositions for the weight tensor in neural network layers
- Investigated how to reduce dataset bias for better machine learning models
- Created a diffusion model that produces images based on spatial layouts and depth values
- Utilized machine unlearning to maintain dataset privacy while removing dataset bias
- Collaborated with other researchers on Test Vector Leakage Assessment for symmetric and asymmetric encryption algorithms
- Created hardware implementation of Elliptic Curve Integrated Encryption Scheme and Elliptic Curve Digital Signature Algorithm with the goal of configurability
- Investigated and implemented mitigation techniques for different types of attacks against the elliptic curve modules

L3Harris Technologies

SOFTWARE ENGINEERING INTERN, SPACE AND AIRBORNE SYSTEMS SEGMENT

Palm Bay, Florida

May 2022 - July 2022

- Implemented C++ classes for message handling and telemetry
- Created unit tests using googletest for implemented classes
- Updated documentation with updated design decisions and implementation details

L3Harris Technologies

SOFTWARE ENGINEERING INTERN, SPACE AND AIRBORNE SYSTEMS SEGMENT

Palm Bay, Florida

May 2021 - July 2021

- Designed software module structure for battery trays on a satellite
- Utilized design patterns in structuring the design
- Established virtual machine environment that comes pre-loaded with all tools required for doing development on the project to cut down on time needed to request all software and tools

L3Harris Technologies

SOFTWARE ENGINEERING INTERN, SPACE AND AIRBORNE SYSTEMS SEGMENT

Palm Bay, Florida

July 2020 - Aug. 2020

- Provided support to final efforts of system worked on previously (May 2019 - August 2019)
- Tested final end system before delivery of product to formal qualification testing
- Ensured testing documents were valid in exact testing procedures and grammar

L3Harris Technologies

Palm Bay, Florida

SOFTWARE ENGINEERING INTERN, SPACE AND AIRBORNE SYSTEMS SEGMENT

May 2019 - Aug. 2019

- Worked on a team responsible for the creation of software using VxWorks and C++ to monitor a space satellite and its hardware
- Built unit tests to test components of the baseline and to prove that requirements were met
- Implemented several requirements into the baseline that were missing or had changed

Teaching Experience

University of Florida Engineering Education Department

Gainesville, Florida

GRADUATE TEACHING ASSISTANT

Aug. 2021 - May 2022

- Assisted students in Computer Engineering Design 1 and 2, a two semester sequence for undergraduate computer engineers to complete their senior capstone
- Graded students' assignments
- Acted as the client for groups to direct them forward to meet course requirements of the project
- Answered students' questions regarding assigned labs, project milestones and deliverables, and their projects.

University of Florida CISE Department

Gainesville, Florida

UNDERGRADUATE TEACHING ASSISTANT

Aug. 2018 - May 2019

- Helped students in Programming Fundamentals 1 (Aug. - Dec. 2018) and Programming Fundamentals 2 (Jan. - May 2019) with questions they had relating to the programming languages and concepts taught
- Answered any questions related to strong programming skills and Java or C++-specific concepts
- Graded students' code for labs, projects, and free responses on exams
- Assisted first time programmers in Programming Fundamentals 1 adjust to the course and they type of work required to succeed in the class

Research

Security IP

Gainesville, Florida

UF EMBEDDED SYSTEMS LAB

Jan. 2022 - Aug. 2023

- Created hardware implementation of Elliptic Curve Integrated Encryption Scheme and Elliptic Curve Digital Signature Algorithm with the goal of configurability
- Investigated and implemented mitigation techniques for different types of attacks against the modules

Amphibian and Lilypad

Gainesville, Florida

UF ENGAGING LEARNING LAB

Jan. 2020 - Apr. 2022

- Amphibian is an IntelliJ plugin that allows for users to go from block-based code representation to traditional text representation
- Implemented exporting of blocks as images and animations to use for reference or educational material
- Conceptualized a frame-based editor called Lilypad that provides block-like syntax highlighting to the traditional text representation

Publications

Journal Articles

- A. Jayasena, E. Andrews, and P. Mishra, "TVLA*: Test Vector Leakage Assessment on Hardware Implementations of Asymmetric Cryptography Algorithms," IEEE Transactions on Very Large Scale Integration (VLSI) Systems, vol. 31, no. 9, pp. 1269–1279, Sep. 2023, doi: 10.1109/TVLSI.2023.3297027.

Conference Proceedings

- E. Andrews, D. Bau, and J. Blanchard, "From Droplet to Lilypad: Present and Future of Dual-Modality Environments," in 2021 IEEE Symposium on Visual Languages and Human-Centric Computing (VL/HCC), Oct. 2021, pp. 1–2. doi: 10.1109/VL/HCC51201.2021.9576355.

Book Chapters

- E. Andrews, Z. Pan, and P. Mishra, "Explainable Artificial Intelligence," in Explainable AI for Cybersecurity, Springer, 2023, pp. 29–51.

Preprints and Other Publications

- E. Andrews and P. Mishra, “Explainable Metric Learning for Deflating Data Bias.” SRC TECHCON, Sep. 2024.
- Z. Pan, E. Andrews, L. Chang, and P. Mishra, “Privacy-Preserving Debiasing using Data Augmentation and Machine Unlearning.” arXiv, Apr. 19, 2024. doi: 10.48550/arXiv.2404.13194.
- E. Andrews, Z. Pan, and P. Mishra, “Towards Accurate Measurement of Pretraining Bias.” SRC TECHCON, Sep. 2023.
- K. Rani, E. Andrews, A. Jayasena and P. Mishra, “Defending Elliptic Curve Cryptography against Laser Fault injection Attacks”, GOMACTech Conference, San Diego, California, Mar. 20-23, 2023.

Awards

Aug. 2024 **L3Harris Endowment Ph.D. Fellow**, Large-Scale Quantum Computing Simulation via Generative AI *Gainesville, Florida*